Publication List

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16	**6**	**2**	**557**	**12**
peer-reviewed papers	as first author	preprints under review	citations	h-index

Citations and h-index from Web of Science (ResearcherID ABD-4954-2021), 12 October 2026.

Peer-reviewed publications

First author

1. **Maarten R. Dobbelaere**, Helen F. Sneddon. The Stubborn Persistence of Toxic Solvents in Chemical Synthesis. *Angewandte Chemie International Edition* **2026**, e1677569. DOI: 10.1002/anie.1677569
   Covered by *Nature* News, *C&EN*, *Spektrum der Wissenschaft*, *Cumhuriyet* and *ScienceNet China*.
2. **Maarten R. Dobbelaere**, István Lengyel, Christian V. Stevens, Kevin M. Van Geem. Geometric deep learning for molecular property predictions with chemical accuracy across chemical space. *Journal of Cheminformatics* **2024**, 16, 99. DOI: 10.1186/s13321-024-00895-0
3. **Maarten R. Dobbelaere**, István Lengyel, Christian V. Stevens, Kevin M. Van Geem. Rxn-INSIGHT: fast chemical reaction analysis using bond-electron matrices. *Journal of Cheminformatics* **2024**, 16, 37. DOI: 10.1186/s13321-024-00834-z
4. **Maarten R. Dobbelaere**, Yannick Ureel, Florence H. Vermeire, Lowie Tomme, Christian V. Stevens, Kevin M. Van Geem. Machine Learning for Physicochemical Property Prediction of Complex Hydrocarbon Mixtures. *Industrial & Engineering Chemistry Research* **2022**, 61(24), 8581–8594. DOI: 10.1021/acs.iecr.2c00442
5. **Maarten R. Dobbelaere**, Pieter P. Plehiers, Ruben Van de Vijver, Christian V. Stevens, Kevin M. Van Geem. Machine Learning in Chemical Engineering: Strengths, Weaknesses, Opportunities, and Threats. *Engineering* **2021**, 7(9), 1201–1211. DOI: 10.1016/j.eng.2021.03.019
6. **Maarten R. Dobbelaere**, Pieter P. Plehiers, Ruben Van de Vijver, Christian V. Stevens, Kevin M. Van Geem. Learning Molecular Representations for Thermochemistry Prediction of Cyclic Hydrocarbons and Oxygenates. *Journal of Physical Chemistry A* **2021**, 125(23), 5166–5179. DOI: 10.1021/acs.jpca.1c01956

Co-author

7. Lowie Tomme, Jeroen Aerssens, István Lengyel, Maarten K. Sabbe, **Maarten R. Dobbelaere**, Christian V. Stevens, Kevin M. Van Geem. Semi-Empirical Evaluation of Hindered Internal Rotors for Accelerated Thermodynamics Predictions. *Journal of Physical Chemistry A* **2026**, 130(10), 2158–2168. DOI: 10.1021/acs.jpca.5c08672
8. Yannick Ureel, **Maarten R. Dobbelaere**, István Lengyel, Maarten K. Sabbe, Kevin M. Van Geem. How can active machine learning aid kinetic model generation, and why should we care? *Engineering* **2025**, 52, 14–18. DOI: 10.1016/j.eng.2025.08.009

9. Caroline T. A. Moermond, Neele Puhlmann, Lowik Pieters, Avtar Matharu, Lieselot Boone, **Maarten R. Dobbelaere**, Héloïse Proquin, Klaus Kümmerer, Ad M. J. Ragas, Rodrigo Vidaurre, Bastiaan J. Venhuis, Delphine De Smedt. Eco-pharma dilemma: navigating environmental sustainability trade-offs within the lifecycle of pharmaceuticals – a comment. *Sustainable Chemistry and Pharmacy* **2025**, 43, 101893. DOI: 10.1016/j.scp.2024.101893
10. Lowie Tomme, Yannick Ureel, **Maarten R. Dobbelaere**, István Lengyel, Florence H. Vermeire, Christian V. Stevens, Kevin M. Van Geem. Machine learning applications for thermochemical and kinetic property prediction. *Reviews in Chemical Engineering* **2025**, 41(4), 419–449. DOI: 10.1515/revce-2024-0027
11. Yasemen Kuddusi, **Maarten R. Dobbelaere**, Kevin M. Van Geem, Andreas Züttel. Accelerated design of nickel-cobalt based catalysts for CO_2 hydrogenation with human-in-the-loop active machine learning. *Catalysis Science & Technology* **2024**, 14(21), 6307–6320. DOI: 10.1039/d4cy00873a
12. Yannick Ureel, **Maarten R. Dobbelaere**, Yi Ouyang, Kevin De Ras, Maarten K. Sabbe, Guy B. Marin, Kevin M. Van Geem. Active machine learning for chemical engineers: a bright future lies ahead! *Engineering* **2023**, 27, 23–30. DOI: 10.1016/j.eng.2023.02.019
13. Onur Dogu, Andreas Eschenbacher, Robin John Varghese, **Maarten R. Dobbelaere**, Dagmar R. D'hooge, Paul H. M. Van Steenberge, Kevin M. Van Geem. Bayesian tuned kinetic Monte Carlo modeling of polystyrene pyrolysis: Unraveling the pathways to its monomer, dimers, and trimers formation. *Chemical Engineering Journal* **2023**, 455, 140708. DOI: 10.1016/j.cej.2022.140708
14. Yannick Ureel, **Maarten R. Dobbelaere**, Oğuzhan Akin, Robin John Varghese, César Pernalete Piña, Joris W. Thybaut, Kevin M. Van Geem. Active learning-based exploration of the catalytic pyrolysis of plastic waste. *Fuel* **2022**, 328, 125340. DOI: 10.1016/j.fuel.2022.125340
15. Yi Ouyang, Laurien A. Vandewalle, Lin Chen, Pieter P. Plehiers, **Maarten R. Dobbelaere**, Geraldine J. Heynderickx, Guy B. Marin, Kevin M. Van Geem. Speeding up turbulent reactive flow simulation via a deep artificial neural network: A methodology study. *Chemical Engineering Journal* **2022**, 429, 132442. DOI: 10.1016/j.cej.2021.132442
16. Pieter P. Plehiers, Connor W. Coley, Hanyu Gao, Florence H. Vermeire, **Maarten R. Dobbelaere**, Christian V. Stevens, Kevin M. Van Geem, William H. Green. Artificial Intelligence for Computer-Aided Synthesis In Flow: Analysis and Selection of Reaction Components. *Frontiers in Chemical Engineering* **2020**, 2, 5. DOI: 10.3389/fceng.2020.00005

Preprints and manuscripts under review

P1. Daniel P. Armstrong, **Maarten R. Dobbelaere**, Valentas Olikauskas, Helena Avila, Octavian Susanu, Jérôme Waser, Philippe Schwaller. Agentic generation of verifiable rules for deterministic, self-expanding reaction classification. *arXiv* **2026**, arXiv:2607.01061. Under review at Nature Machine Intelligence.

P2. Daniel Armstrong, Xuan-Vu Nguyen, Octavian Susanu, Gabriel Gibberd, Théo A. Neukomm, Taddäus Strunden, Dan Forster, Morgane Delattre, Shawn Teh, Clément Rols, John Federice, Hayden Leatherwood, M. Lavelle Barnes, **Maarten R. Dobbelaere**, Peter Wipf, Jon T. Njardarson, Jieping Zhu, Philippe Schwaller. Strategy-first synthesis planning for complex natural products. *arXiv* **2026**, arXiv:2608.07454. Under review at Nature.